Huawei Access Node Hardware/Environmental Fault Resolution Standard Operating Procedure (SOP) for YEL-ACC-17 at Yelahanka Sub-station

Node: YEL-ACC-17 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_7592

Description

This failure domain occurs on the Huawei Access Node due to Hardware/Environmental Faults (ASIC memory parity errors, line card crashes, and power supply redundancy loss) in the YEL-ACC-17 node at Yelahanka Sub-station. The root cause can be attributed to the failure of the redundant power supply system or the line card's inability to handle the increased traffic, leading to a crash. This Hardware/Environmental Fault is a critical issue that requires immediate attention to prevent network downtime and ensure the reliability of the network. The Huawei Access Node's architecture, featuring a distributed architecture with multiple line cards, enables it to provide high availability, but it also increases the complexity of the system. The fault-tolerant design ensures that the node can continue to operate even in the event of a hardware failure, but the ASIC memory parity errors can still occur due to the high traffic and high-speed data transmission. The line card crashes can be caused by the overload of traffic or the malfunction of the line card's hardware components. The power supply redundancy loss is a critical issue that can cause the entire node to fail, resulting in network downtime. Therefore, it is essential to identify and troubleshoot the root cause of the failure promptly to prevent network downtime and ensure the reliability of the network.

Pre-requisite Checks

- Verify that the redundant power supply system is functioning correctly.

- Check the status of the line cards and ensure that they are functioning within normal operating parameters.

- Verify that the network traffic is within the recommended limits for the Huawei Access Node.

- Ensure that the remote hands are dispatched to Yelahanka Sub-station to facilitate the investigation and resolution process.

Potential Root Causes

Cause	Probability
ASIC memory parity error due to high traffic and high-speed data transmission.	High
Line card crash due to overload of traffic or malfunction of hardware components.	Medium
Power supply redundancy loss.	Low

Diagnostic Commands & Expected Outputs

```
Command: display power supply, Expected Output: Power supply status: Online/Offline, Description: This command displays the status of the power supply system, enabling the identification of any potential issues with the power supply redundancy.

Command: display line-card power supply, Expected Output: Line card power supply status: Online/Offline, Description: This command displays the status of the power supply system for each line card, enabling the identification of any potential issues with the line card's power supply.

Command: display system-asic-status, Expected Output: ASIC status: Normal/Abnormal, Description: This command displays the status of the ASIC, enabling the identification of any potential issues with the ASIC.
```

Resolution Steps

Step 1: Step: Replace the faulty power supply unit (PSU)., Command: replace power supply unit 1, Description: This command replaces the faulty PSU with a new one, ensuring that the power supply redundancy is restored, and the node can continue to operate normally.

Step 2: Step: Replace the faulty line card., Command: replace line card 1, Description: This command replaces the faulty line card with a new one, ensuring that the line card's power supply and hardware components are restored to normal operating parameters.

Step 3: Step: Clear the ASIC memory parity error., Command: clear system-asic-status, Description: This command clears the ASIC memory parity error, ensuring that the ASIC can continue to operate normally.

Post-Check Verification

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Command: display power supply, Expected Output: Power supply status: Online/Online, Description: This command verifies that the power supply redundancy is restored and functioning correctly.

Command: display line-card power supply, Expected Output: Line card power supply status: Online/Online, Description: This command verifies that the line card's power supply is restored to normal operating parameters.

Command: display system-asic-status, Expected Output: ASIC status: Normal/Normal, Description: This command verifies that the ASIC is operating normally and there are no memory parity errors.
```

Roll-back Procedures

- Step: Restore the original configuration., Command: undo replace power supply unit 1, Description: This command restores the original configuration, ensuring that the node returns to its previous state.

- Step: Restore the original configuration., Command: undo replace line card 1, Description: This command restores the original configuration, ensuring that the line card returns to its previous state.

- Step: Restore the original configuration., Command: undo clear system-asic-status, Description: This command restores the original configuration, ensuring that the ASIC returns to its previous state.